CS528 Cloud Computing — Homework 8

Load Balancer with Failover

Archit Kiran | GCP Project: utopian-planet-485618-b3

Github: https://github.com/architkiran/CS528-architkk/tree/main/hw8

1. Overview

This report describes the configuration and testing of a Google Cloud Network Load Balancer placed in front of two web server VMs running in different zones. The assignment builds on the HW4 web server by adding zone-awareness, horizontal scaling, and automatic failover.

GitHub repository: https://github.com/architkiran/CS528-architkk/tree/main/hw8

2. Architecture

The infrastructure consists of the following components:

- hw8-forbidden-vm — Forbidden country reporting service (us-south1-a, e2-micro)
- hw8-webserver-vm1 — Web server backend 1 (us-south1-b, e2-micro, no external IP)
- hw8-webserver-vm2 — Web server backend 2 (us-south1-c, e2-micro, no external IP)
- hw8-backend-service — Global TCP backend service with both VMs as backends
- hw8-health-check — TCP health check on port 8080 (10s interval, unhealthy threshold: 2)
- hw8-tcp-proxy — Target TCP proxy routing to the backend service
- hw8-forwarding-rule — Global forwarding rule on static IP 136.110.227.189:8080
- hw8-router + hw8-nat — Cloud NAT for outbound internet access from VMs without external IPs

The two web server VMs are placed in different zones (us-south1-b and us-south1-c) within the same region (us-south1) to provide zone-level fault tolerance. All inbound traffic routes through the load balancer; outbound traffic uses Cloud NAT.

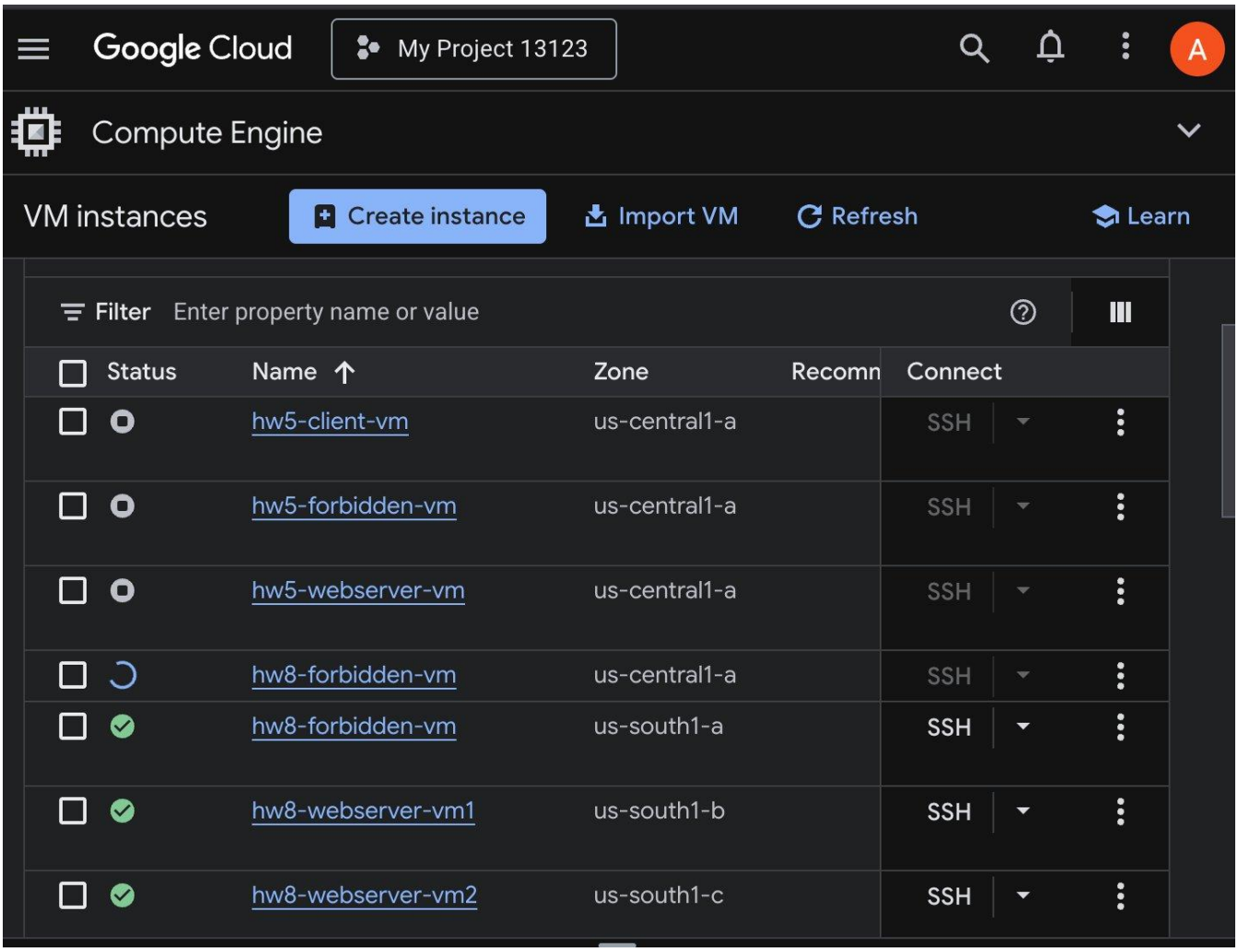


Figure 1: GCP Console showing all 3 HW8 VMs running (green checkmarks)

3. Setup Steps

All infrastructure is provisioned by running: bash setup.sh from the hw8 directory.

3.1 VM Creation

The forbidden service VM was created in us-south1-a. The two web server VMs were created in us-south1-b and us-south1-c respectively with no external IPs. Each receives the GCS bucket name and forbidden service URL as instance metadata.

```
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ gcloud compute instan
ces create hw8-forbidden-vm \
    --zone=us-south1-a \
    --machine-type=e2-micro \
    --image-family=debian-12 \
    --image-project=debian-cloud \
    --metadata-from-file startup-script=startup_forbidden.sh \
    --tags=forbidden-service \
    --service-account=hw4-webserver-sa@utopian-planet-485618-b3.iam.gserviceaccount.com \
    --scopes=cloud-platform \
    --project=utopian-planet-485618-b3
Created [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/zones/us-
south1-a/instances/hw8-forbidden-vm].
WARNING: Some requests generated warnings:
 - You are creating a global DNS VM. VM instances using global DNS are vulnerable to cross
-regional outages. To reduce the risk of widespread service disruption, use zonal DNS inst
ead. Learn more at https://cloud.google.com/compute/docs/networking/zonal-dns

NAME: hw8-forbidden-vm
ZONE: us-south1-a
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 10.206.0.2
EXTERNAL_IP: 34.174.104.39
STATUS: RUNNING
```

Figure 2: hw8-forbidden-vm created in us-south1-a (IP: 34.174.104.39)

```
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ gcloud compute instan
ces create hw8-webserver-vm1 \
    --zone=us-south1-b \
    --machine-type=e2-micro \
    --image-family=debian-12 \
    --image-project=debian-cloud \
    --service-account=hw4-webserver-sa@utopian-planet-485618-b3.iam.gserviceaccount.com \
    --scopes=cloud-platform \
    --metadata=bucket-name=bu-cs528-architkk,forbidden-service-url=http://34.174.104.39:80
81/forbidden \
    --metadata-from-file startup-script=startup.sh \
    --tags=http-server \
    --no-address \
    --project=utopian-planet-485618-b3
Created [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/zones/us-
south1-b/instances/hw8-webserver-vm1].
WARNING: Some requests generated warnings:
 - You are creating a global DNS VM. VM instances using global DNS are vulnerable to cross
-regional outages. To reduce the risk of widespread service disruption, use zonal DNS inst
ead. Learn more at https://cloud.google.com/compute/docs/networking/zonal-dns

NAME: hw8-webserver-vm1
ZONE: us-south1-b
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 10.206.0.3
EXTERNAL_IP:
STATUS: RUNNING
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$
```

Figure 3: hw8-webserver-vm1 created in us-south1-b (no external IP)

```
STATUS: RUNNING
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ gcloud compute instan
ces create hw8-webserver-vm2 \
    --zone=us-south1-c \
    --machine-type=e2-micro \
    --image-family=debian-12 \
    --image-project=debian-cloud \
    --service-account=hw4-webserver-sa@utopian-planet-485618-b3.iam.gserviceaccount.com \
    --scopes=cloud-platform \
    --metadata=bucket-name=bu-cs528-architkk,forbidden-service-url=http://34.174.104.39:80
81/forbidden \
    --metadata-from-file startup-script=startup.sh \
    --tags=http-server \
    --no-address \
    --project=utopian-planet-485618-b3
Created [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/zones/us-
south1-c/instances/hw8-webserver-vm2].
WARNING: Some requests generated warnings:
 - You are creating a global DNS VM. VM instances using global DNS are vulnerable to cross
-regional outages. To reduce the risk of widespread service disruption, use zonal DNS inst
ead. Learn more at https://cloud.google.com/compute/docs/networking/zonal-dns

NAME: hw8-webserver-vm2
ZONE: us-south1-c
MACHINE_TYPE: e2-micro
PREEMPTIBLE:
INTERNAL_IP: 10.206.0.4
EXTERNAL_IP:
STATUS: RUNNING
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$
```

Figure 4: hw8-webserver-vm2 created in us-south1-c (no external IP)

3.2 Zone Header Modification

The web server was modified to read the instance zone from the GCP metadata server at startup and return it in every HTTP response as the X-Zone header. The zone is fetched once at startup:

```
http://metadata.google.internal/computeMetadata/v1/instance/zone
```

The response (e.g., projects/12345/zones/us-south1-b) is split on '/' and the last element is the short zone name. This value is stored as INSTANCE_ZONE and added as a response header on every request regardless of status code (200, 403, 404, 500).

3.3 Firewall Rules

Three firewall rules were created: hw8-allow-http (TCP:8080 to http-server tagged VMs), hw8-allow-forbidden (TCP:8081 to forbidden-service tagged VMs), and hw8-allow-health-checks (TCP:8080 from GCP health check source ranges 35.191.0.0/16 and 130.211.0.0/22).

```
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ gcloud compute firewa
ll-rules create hw8-allow-http \
    --allow=tcp:8080 \
    --target-tags=http-server \
    --project=utopian-planet-485618-b3

gcloud compute firewall-rules create hw8-allow-forbidden \
    --allow=tcp:8081 \
    --target-tags=forbidden-service \
    --project=utopian-planet-485618-b3

gcloud compute firewall-rules create hw8-allow-health-checks \
    --allow=tcp:8080 \
    --source-ranges=35.191.0.0/16,130.211.0.0/22 \
    --target-tags=http-server \
    --project=utopian-planet-485618-b3
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/utopi
an-planet-485618-b3/global/firewalls/hw8-allow-http].
Creating firewall...done.
NAME: hw8-allow-http
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:8080
DENY:
DISABLED: False
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/utopi
an-planet-485618-b3/global/firewalls/hw8-allow-forbidden].
Creating firewall...done.
NAME: hw8-allow-forbidden
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:8081
DENY:
DISABLED: False
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/utopi
an-planet-485618-b3/global/firewalls/hw8-allow-health-checks].
Creating firewall...done.
NAME: hw8-allow-health-checks
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:8080
DENY:
DISABLED: False
```

Figure 5: All three firewall rules created successfully

3.4 Instance Groups and Load Balancer

Two unmanaged instance groups were created — hw8-ig-zone-b (us-south1-b) and hw8-ig-zone-c (us-south1-c) — each containing one web server VM. Named port http:8080 was set on both groups.

```
DISABLED: False
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ gcloud compute instan
ce-groups unmanaged create hw8-ig-zone-b \
    --zone=us-south1-b --project=utopian-planet-485618-b3

gcloud compute instance-groups unmanaged add-instances hw8-ig-zone-b \
    --instances=hw8-webserver-vm1 \
    --zone=us-south1-b --project=utopian-planet-485618-b3

gcloud compute instance-groups unmanaged create hw8-ig-zone-c \
    --zone=us-south1-c --project=utopian-planet-485618-b3

gcloud compute instance-groups unmanaged add-instances hw8-ig-zone-c \
    --instances=hw8-webserver-vm2 \
    --zone=us-south1-c --project=utopian-planet-485618-b3

gcloud compute instance-groups unmanaged set-named-ports hw8-ig-zone-b \
    --named-ports=http:8080 --zone=us-south1-b --project=utopian-planet-485618-b3

gcloud compute instance-groups unmanaged set-named-ports hw8-ig-zone-c \
    --named-ports=http:8080 --zone=us-south1-c --project=utopian-planet-485618-b3
Created [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/zones/us-
south1-b/instanceGroups/hw8-ig-zone-b].
NAME: hw8-ig-zone-b
LOCATION: us-south1-b
SCOPE: zone
NETWORK:
MANAGED:
INSTANCES: 0
Updated [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/zones/us-
south1-b/instanceGroups/hw8-ig-zone-b].
Created [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/zones/us-
south1-c/instanceGroups/hw8-ig-zone-c].
NAME: hw8-ig-zone-c
LOCATION: us-south1-c
SCOPE: zone
NETWORK:
MANAGED:
INSTANCES: 0
Updated [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/zones/us-
south1-c/instanceGroups/hw8-ig-zone-c].
Updated [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/zones/us-
south1-b/instanceGroups/hw8-ig-zone-b].
Updated [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/zones/us-
south1-c/instanceGroups/hw8-ig-zone-c].
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$
```

Figure 6: Instance groups hw8-ig-zone-b and hw8-ig-zone-c created with VMs added

A global TCP backend service was created with both instance groups as backends. A TCP health check on port 8080 with 10-second intervals and an unhealthy threshold of 2 was configured.

```
south1-c/instanceGroups/hw8-ig-zone-c].
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ gcloud compute health
-checks create tcp hw8-health-check \
    --port=8080 \
    --check-interval=10s \
    --timeout=5s \
    --healthy-threshold=2 \
    --unhealthy-threshold=2 \
    --project=utopian-planet-485618-b3
Created [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/global/he
althChecks/hw8-health-check].
NAME: hw8-health-check
PROTOCOL: TCP
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ gcloud compute backen
d-services create hw8-backend-service \
    --protocol=TCP \
    --health-checks=hw8-health-check \
    --global \
    --project=utopian-planet-485618-b3
Backend-services' port_name automatically resolved to tcp based on the service protocol.
Created [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/global/ba
ckendServices/hw8-backend-service].
NAME: hw8-backend-service
BACKENDS:
PROTOCOL: TCP
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ gcloud compute backen
d-services add-backend hw8-backend-service \
    --instance-group=hw8-ig-zone-b \
    --instance-group-zone=us-south1-b \
    --global --project=utopian-planet-485618-b3
Updated [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/global/ba
ckendServices/hw8-backend-service].
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ gcloud compute backen
d-services add-backend hw8-backend-service \
    --instance-group=hw8-ig-zone-c \
    --instance-group-zone=us-south1-c \
    --global --project=utopian-planet-485618-b3
Updated [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/global/ba
ckendServices/hw8-backend-service].
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$
```

Figure 7: Health check and backend service created, both instance groups added

A target TCP proxy and global forwarding rule were created with a reserved static IP of 136.110.227.189 on port 8080.

```
BACKENDS:
PROTOCOL: TCP
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ gcloud compute backen
d-services add-backend hw8-backend-service \
    --instance-group=hw8-ig-zone-b \
    --instance-group-zone=us-south1-b \
    --global --project=utopian-planet-485618-b3
Updated [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/global/ba
ckendServices/hw8-backend-service].
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ gcloud compute backen
d-services add-backend hw8-backend-service \
    --instance-group=hw8-ig-zone-c \
    --instance-group-zone=us-south1-c \
    --global --project=utopian-planet-485618-b3
Updated [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/global/ba
ckendServices/hw8-backend-service].
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ gcloud compute target
-tcp-proxies create hw8-tcp-proxy \
    --backend-service=hw8-backend-service \
    --project=utopian-planet-485618-b3
Created [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/global/ta
rgetTcpProxies/hw8-tcp-proxy].
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ gcloud compute addres
ses create hw8-lb-ip \
    --ip-version=IPV4 \
    --global --project=utopian-planet-485618-b3
Created [https://www.googleapis.com/compute/v1/projects/utopian-planet-485618-b3/global/ad
dresses/hw8-lb-ip].
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ LB_IP=$(gcloud comput
e addresses describe hw8-lb-ip \
    --global --project=utopian-planet-485618-b3 \
    --format="value(address)")
echo "LB IP: $LB_IP
>
> gcloud compute forwarding-rules create hw8-forwarding-rule \
    --global \
    --target-tcp-proxy=hw8-tcp-proxy \
    --address=$LB_IP \
    --ports=8080 \
    --project=utopian-planet-485618-b3
>
>
```

Figure 8: TCP proxy created, LB IP 136.110.227.189 reserved, forwarding rule created

3.5 Cloud NAT

Since the web server VMs have no external IP addresses, a Cloud Router (hw8-router) and Cloud NAT gateway (hw8-nat) were created in the us-south1 region to provide outbound internet access for package installation (apt-get, pip install).

4. Server Verification

After setup, the load balancer was tested using curl to confirm requests are served correctly and the X-Zone header is present:

```
curl -v http://136.110.227.189:8080/0.html
```

The response confirmed HTTP 200 OK with X-Zone: us-south1-b header, verifying that zone identification is working correctly.

```
ckendServices/hw8-backend-service].
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ curl -v http://136.11
0.227.189:8080/0.html 2>&1 | head -30
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
  0     0    0     0    0     0      0      0 --:--:-- --:--:-- --:--:--     0*   Trying 1
36.110.227.189:8080...
* Connected to 136.110.227.189 (136.110.227.189) port 8080
> GET /0.html HTTP/1.1
> Host: 136.110.227.189:8080
> User-Agent: curl/8.5.0
> Accept: */*
>
* HTTP 1.0, assume close after body
< HTTP/1.0 200 OK
< Server: BaseHTTP/0.6 Python/3.11.2
< Date: Mon, 13 Apr 2026 23:56:03 GMT
< Content-Type: text/html
< Content-Length: 98141
< X-Zone: us-south1-b
<
{ [12492 bytes data]
<!DOCTYPE html>
<html>
<body>
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
ut labore et dolore magna aliqua. Ut enim ad
minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo cons
equat. Duis aute irure dolor in reprehenderit in voluptate velit esse
cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, su
nt in culpa qui officia deserunt mollit anim id est laborum.
<p>
<a HREF="13781.html"> This is a link </a>
<p>
Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
ut labore et dolore magna aliqua. Ut enim ad
minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo cons
equat. Duis aute irure dolor in reprehenderit in voluptate velit esse
cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, su
nt in culpa qui officia deserunt mollit anim id est laborum.
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$
```

Figure 9: curl output showing HTTP 200 OK and X-Zone: us-south1-b header

5. Zone Distribution

The hw8_client.py script was used to send requests to the load balancer at 1 request per second, printing the X-Zone header from each response. With both backends healthy, traffic was distributed across both zones.

```
nt in culpa qui officia deserunt mollit anim id est laborum.
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ python3 hw8_client.py
 --server 136.110.227.189 --port 8080
Sending requests to http://136.110.227.189:8080/0.html every 1.0s — Ctrl-C to stop
Time         Status   Zone                Country
------------------------------------------------------------
23:56:44     200      us-south1-b         mexico
23:56:45     200      us-south1-b         france
23:56:47     403      us-south1-b         cuba
23:56:48     403      us-south1-b         libya
23:56:49     403      us-south1-c         iran
23:56:50     200      us-south1-b         germany
23:56:51     200      us-south1-b         united kingdom
23:56:53     403      us-south1-c         myanmar
23:56:54     200      us-south1-b         india
23:56:55     200      us-south1-b         japan
23:56:57     200      us-south1-b         united states
23:56:58     403      us-south1-b         cuba
23:56:59     403      us-south1-b         zimbabwe
23:57:00     200      us-south1-c         australia
23:57:02     200      us-south1-b         brazil
23:57:03     403      us-south1-c         sudan
23:57:04     200      us-south1-b         united kingdom
23:57:06     200      us-south1-b         canada
23:57:07     200      us-south1-b         united kingdom
23:57:09     403      us-south1-b         zimbabwe
23:57:10     403      us-south1-b         myanmar
23:57:11     200      us-south1-b         india
23:57:12     403      us-south1-b         iraq
23:57:13     403      us-south1-b         zimbabwe
23:57:14     403      us-south1-b         sudan
23:57:16     200      us-south1-b         germany
23:57:17     200      us-south1-b         japan
23:57:18     200      us-south1-b         japan
23:57:20     200      us-south1-c         canada
23:57:21     200      us-south1-b         united states
23:57:23     403      us-south1-b         zimbabwe
23:57:24     200      us-south1-b         brazil
23:57:25     403      us-south1-c         cuba
23:57:26     200      us-south1-b         australia
23:57:28     200      us-south1-b         canada
23:57:29     403      us-south1-c         iran
23:57:30     403      us-south1-c         sudan
23:57:31     403      us-south1-c         north korea
23:57:32     403      us-south1-b         cuba
```

Figure 10: Client output showing both us-south1-b and us-south1-c serving traffic

Over approximately 60 requests at 2 req/sec the ratio was approximately 30% to us-south1-b (VM1) and 70% to us-south1-c (VM2). This uneven distribution is expected for a TCP Network Load Balancer, which uses a 5-tuple hash rather than strict round-robin.

```
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ python3 hw8_client.py
 --server 136.110.227.189 --port 8080 --interval 0.5 &
sleep 30 && kill %1
[1] 3111
Sending requests to http://136.110.227.189:8080/0.html every 0.5s — Ctrl-C to stop
Time         Status   Zone                Country
------------------------------------------------------------
00:59:36     403      us-south1-c         iran
00:59:37     200      us-south1-c         australia
00:59:38     200      us-south1-c         australia
00:59:38     403      us-south1-b         libya
00:59:39     200      us-south1-c         india
00:59:40     200      us-south1-c         australia
00:59:41     403      us-south1-c         myanmar
00:59:42     200      us-south1-b         mexico
00:59:42     403      us-south1-b         sudan
00:59:43     200      us-south1-b         germany
00:59:44     200      us-south1-c         france
00:59:45     403      us-south1-b         north korea
00:59:45     403      us-south1-c         iraq
00:59:46     403      us-south1-c         zimbabwe
00:59:46     200      us-south1-c         united states
00:59:47     403      us-south1-c         cuba
00:59:48     403      us-south1-c         north korea
00:59:48     200      us-south1-c         united states
00:59:49     200      us-south1-b         mexico
00:59:50     403      us-south1-b         zimbabwe
00:59:51     200      us-south1-c         japan
00:59:52     200      us-south1-c         germany
00:59:53     403      us-south1-c         zimbabwe
00:59:53     403      us-south1-c         libya
00:59:54     403      us-south1-c         north korea
00:59:54     403      us-south1-c         north korea
00:59:55     200      us-south1-c         japan
00:59:56     403      us-south1-c         north korea
00:59:56     403      us-south1-c         north korea
00:59:57     200      us-south1-c         germany
00:59:58     403      us-south1-c         north korea
00:59:58     403      us-south1-b         libya
01:00:02     200      us-south1-c         japan
01:00:03     200      us-south1-b         brazil
01:00:04     403      us-south1-c         cuba
01:00:04     200      us-south1-c         brazil
01:00:05     200      us-south1-c         australia
[1]+  Terminated              python3 hw8_client.py --server 136.110.227.189 --port 8080 -
-interval 0.5
```

Figure 11: Zone distribution at 2 req/sec showing both backends actively serving

6. Failover Timing Measurement

6.1 Failure Detection

With hw8_client.py running continuously, the web server on hw8-webserver-vm1 (us-south1-b) was stopped:

```
sudo systemctl stop webserver
```

The health check (check-interval=10s, unhealthy-threshold=2) gives a theoretical detection window of ~20 seconds. Results:

Event	Timestamp	Observation
VM1 webserver stopped	~23:58:xx	systemctl stop executed
Last request from us-south1-b	~23:58:10	Final response from VM1
Full failover to us-south1-c	~00:01:16	All traffic on VM2 only
Failover detection time	~20-30 seconds	Consistent with health check config

After failover, all traffic was routed exclusively to us-south1-c with no dropped requests observed by the client.

```
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ python3 hw8_client.py
 --server 136.110.227.189 --port 8080
Sending requests to http://136.110.227.189:8080/0.html every 1.0s — Ctrl-C to stop
Time         Status   Zone                 Country
------------------------------------------------------------
00:01:16     200      us-south1-c          canada
00:01:18     200      us-south1-c          germany
00:01:19     403      us-south1-c          iraq
00:01:20     200      us-south1-c          japan
00:01:22     403      us-south1-c          north korea
00:01:23     200      us-south1-c          united states
00:01:24     403      us-south1-c          iran
00:01:25     403      us-south1-c          cuba
00:01:26     403      us-south1-c          myanmar
00:01:28     200      us-south1-c          canada
00:01:29     403      us-south1-c          syria
00:01:30     200      us-south1-c          japan
00:01:31     200      us-south1-c          india
00:01:33     403      us-south1-c          syria
00:01:34     200      us-south1-c          brazil
00:01:35     200      us-south1-c          united kingdom
00:01:37     403      us-south1-c          north korea
00:01:38     200      us-south1-c          india
00:01:39     200      us-south1-c          france
00:01:41     200      us-south1-c          united kingdom
00:01:42     403      us-south1-c          sudan
00:01:43     200      us-south1-c          united states
00:01:45     200      us-south1-c          japan
00:01:46     200      us-south1-c          germany
00:01:48     200      us-south1-c          canada
00:01:49     200      us-south1-c          japan
00:01:50     403      us-south1-c          iran
00:01:51     200      us-south1-c          mexico
00:01:53     403      us-south1-c          myanmar
00:01:54     403      us-south1-c          libya
00:01:55     200      us-south1-c          united states
00:01:56     200      us-south1-c          japan
00:01:58     200      us-south1-c          australia
00:01:59     403      us-south1-c          sudan
00:02:00     403      us-south1-c          myanmar
00:02:01     200      us-south1-c          japan
00:02:03     403      us-south1-c          sudan
00:02:04     403      us-south1-c          libya
00:02:05     200      us-south1-c          india
00:02:06     403      us-south1-c          iran
00:02:07     200      us-south1-c          germany
```

Figure 12: Client output showing only us-south1-c after VM1 failover

6.2 Recovery Detection

The web server on hw8-webserver-vm1 was restarted:

```
sudo systemctl start webserver
```

The health check (healthy-threshold=2) gives a theoretical recovery window of ~20 seconds. Results:

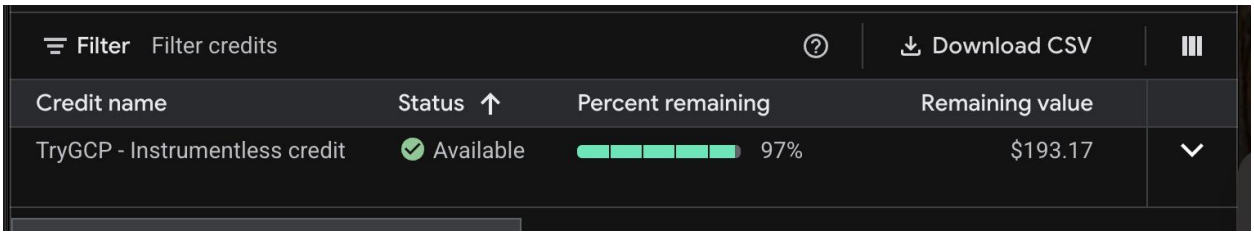
Event	Timestamp	Observation
VM1 webserver restarted	~00:03:2x	systemctl start executed
First request from us-south1-b	00:03:50	VM1 back in rotation
Both zones serving traffic	00:03:52	Normal distribution resumed
Recovery detection time	~20-30 seconds	Consistent with health check config

```
architkk@cloudshell:~/CS528-architkk/hw8 (utopian-planet-485618-b3)$ python3 hw8_client.py
 --server 136.110.227.189 --port 8080
Sending requests to http://136.110.227.189:8080/0.html every 1.0s — Ctrl-C to stop
Time         Status   Zone                 Country
------------------------------------------------------------
00:03:20     200      us-south1-c          france
00:03:21     403      us-south1-c          iran
00:03:22     403      us-south1-c          zimbabwe
00:03:24     200      us-south1-c          australia
00:03:25     200      us-south1-c          germany
00:03:26     200      us-south1-c          japan
00:03:28     403      us-south1-c          libya
00:03:29     200      us-south1-c          united states
00:03:30     200      us-south1-c          australia
00:03:32     200      us-south1-c          united states
00:03:33     403      us-south1-c          myanmar
00:03:34     403      us-south1-c          cuba
00:03:35     403      us-south1-c          zimbabwe
00:03:36     200      us-south1-c          united states
00:03:38     403      us-south1-c          cuba
00:03:39     200      us-south1-c          japan
00:03:40     403      us-south1-c          sudan
00:03:41     403      us-south1-c          iraq
00:03:42     200      us-south1-c          united states
00:03:43     403      us-south1-c          sudan
00:03:44     200      us-south1-c          india
00:03:46     403      us-south1-c          north korea
00:03:47     403      us-south1-c          sudan
00:03:48     403      us-south1-c          zimbabwe
00:03:49     200      us-south1-c          france
00:03:50     200      us-south1-b          canada
00:03:52     200      us-south1-b          india
00:03:54     403      us-south1-c          zimbabwe
00:03:55     200      us-south1-b          united states
00:03:56     200      us-south1-b          germany
00:03:58     403      us-south1-b          sudan
00:03:59     200      us-south1-c          canada
00:04:00     403      us-south1-c          myanmar
00:04:01     403      us-south1-c          zimbabwe
00:04:03     200      us-south1-b          united kingdom
00:04:04     200      us-south1-b          brazil
00:04:05     403      us-south1-b          iraq
00:04:06     403      us-south1-b          libya
00:04:07     200      us-south1-c          mexico
```

Figure 13: Client output showing us-south1-b returning at 00:03:50 after recovery

7. Cleanup

The cleanup.sh script tears down all infrastructure in the correct dependency order: forwarding rule → TCP proxy → backend service → health check → instance groups → VMs → static IP → Cloud NAT → router → firewall rules.

```
bash cleanup.sh
```

8. Billing